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% =====
% 5G NR Link-Level Transceiver Simulation
% EE Communication Systems Homework - Part II Simulation
% =====

clear; clc; close all;

% Set basic system parameters
M = 64; % 64-QAM modulation
k = log2(M); % 6 bits per symbol
numBits = 6e5; % Bit stream length for good BER resolution
snr_dB = 0:2:22; % SNR range for testing
ber_sim = zeros(size(snr_dB));

% Generate input data bits (Source)
dataIn = randi([0 1], numBits, 1);

% Pack bits into symbol values before modulation (Encoding step)
dataSymbolsIn = bit2int(dataIn, k);

% Map integers to 64-QAM complex constellation points
txSymbols = qammod(dataSymbolsIn, M, 'UnitAveragePower', true);

% Run through the channel at different noise levels
for idx = 1:length(snr_dB)

    % Pass through AWGN channel
    rxSymbols = awgn(txSymbols, snr_dB(idx), 'measured');

    % Demodulate received symbols using nearest neighbor decision
    dataSymbolsOut = qamdemod(rxSymbols, M, 'UnitAveragePower', true);

    % Unpack integer symbols back into bit vector
    dataOut = int2bit(dataSymbolsOut, k);

    % Count errors and compute BER
    [numErrors, ber_sim(idx)] = biterr(dataIn, dataOut);
end

% Get theoretical BER values for comparison
ber_theory = berawgn(snr_dB, 'qam', M);

% -----
% Plot 1: BER vs SNR Curve
% -----
figure('Name', '5G NR Link Performance', 'Color', [1 1 1]);
semilogy(snr_dB, ber_sim, 'bo-', 'LineWidth', 2, 'MarkerSize', 6); hold on;
semilogy(snr_dB, ber_theory, 'r--', 'LineWidth', 1.5);
grid on;

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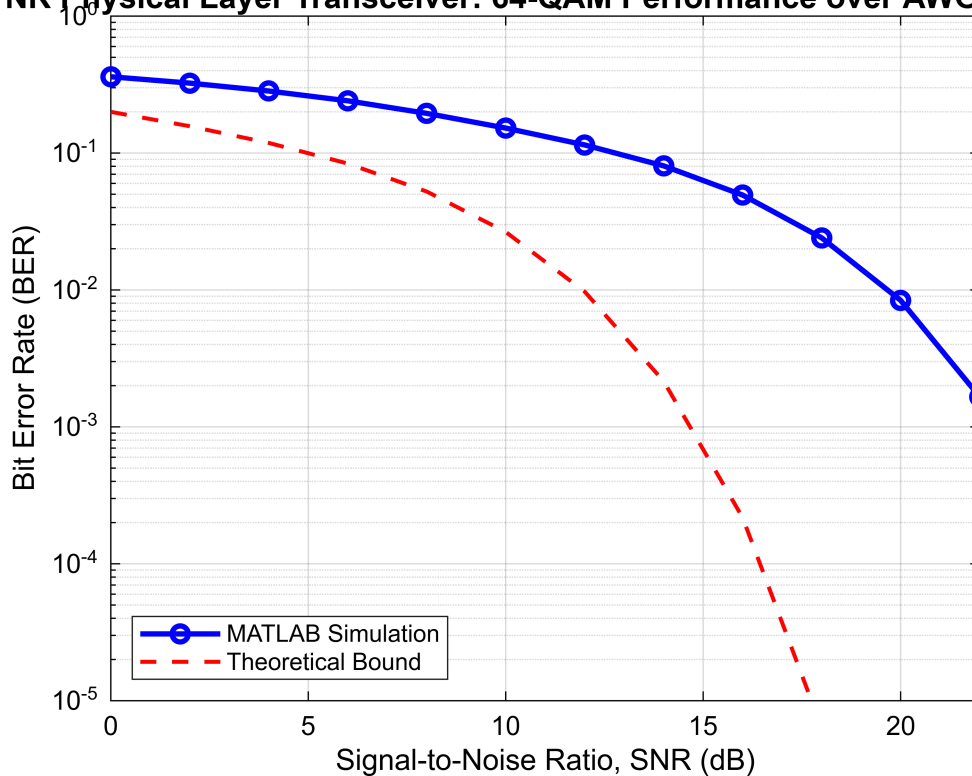
% Labeling with high-contrast text color
xlabel('Signal-to-Noise Ratio, SNR (dB)', 'FontSize', 11, 'Color', [0 0 0]);
ylabel('Bit Error Rate (BER)', 'FontSize', 11, 'Color', [0 0 0]);
title('5G NR Physical Layer Transceiver: 64-QAM Performance over AWGN Channel', ...
      'FontSize', 12, 'Color', [0 0 0]);

% Customizing axes and legend text colors
ax1 = gca;
ax1.XColor = [0 0 0];
ax1.YColor = [0 0 0];

lgd = legend('MATLAB Simulation', 'Theoretical Bound', 'Location',
            'southwest');
set(lgd, 'TextColor', [0 0 0]);
axis([min(snr_dB) max(snr_dB) 1e-5 1]);

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5G NR Physical Layer Transceiver: 64-QAM Performance over AWGN Channel



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% -----
% Plot 2: Constellation Scatter Plot
% -----
figure('Name', 'Constellation Scatter Plot', 'Color', [1 1 1]);

% Subplot 1: Clean Tx Constellation
subplot(1,2,1);
plot(real(txSymbols(1:2000)), imag(txSymbols(1:2000)), 'r.');
grid on;

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title('Transmitted 64-QAM Constellation', 'Color', [0 0 0]);
xlabel('In-Phase (I)', 'Color', [0 0 0]);
ylabel('Quadrature (Q)', 'Color', [0 0 0]);
axis([-1.5 1.5 -1.5 1.5]);
ax2 = gca;
ax2.XColor = [0 0 0];
ax2.YColor = [0 0 0];

% Subplot 2: Noisy Rx Constellation
subplot(1,2,2);
rxSample = awgn(txSymbols(1:2000), 16, 'measured'); % Channel noise at 16 dB
SNR
plot(real(rxSample), imag(rxSample), 'b.');
grid on;
title('Received Constellation at SNR = 16 dB', 'Color', [0 0 0]);
xlabel('In-Phase (I)', 'Color', [0 0 0]);
ylabel('Quadrature (Q)', 'Color', [0 0 0]);
axis([-1.5 1.5 -1.5 1.5]);
ax3 = gca;
ax3.XColor = [0 0 0];
ax3.YColor = [0 0 0];

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